WIANA User Guide

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1. Introduction

Noncovalent interactions are an important type of interaction in compounds in addition to covalent bonds. They generally do not involve the sharing of electron pairs. Compared to strong chemical bonds (usually several hundred kJ/mol), noncovalent interactions are weaker (typically a few to tens of kJ/mol) and have longer interaction distances. Common noncovalent interactions include hydrogen bonds, van der Waals interactions, halogen bonds, chalcogen bonds, pnictogen bonds, π-π stacking interactions, steric hindrance effects, etc. Although individual noncovalent interactions are weak, they play important roles in many fields such as catalysis, synthesis, luminescence, crystal engineering, and drug design through cooperative effects.

Understanding the electronic structure of a system is the basis for elucidating its chemical bonding properties, reactivity, and various physicochemical behaviors. In real space, charge density as a core physical quantity in quantum chemistry, contains information about the ground state of the system. To obtain meaningful chemical information from complex electron density distributions, researchers have developed various real-space function analysis methods. In recent years, real-space function analysis methods based on charge density and its derivatives have become important tools for analyzing interactions. With these methods, researchers can directly observe the specific locations of interactions, evaluate their relative strengths, and deepen understanding of their physical nature. Such methods are not only widely used but also complement traditional numerical calculations. The Reduced Density Gradient (RDG) function [1] is a classical method for analyzing noncovalent interactions (NCIs). It identifies weak interactions through regions of low charge density and low density gradient, and can clearly show their spatial distribution using isosurfaces. The Density Overlap Regions Indicator (DORI) [2] is a function that can simultaneously characterize covalent and noncovalent interaction regions. Its definition requires the calculation of the second derivative of charge density. In addition, the Interaction Region Indicator (IRI) function [3] can also simultaneously characterize covalent and noncovalent interaction regions. It modifies the RDG formula and introduces an empirical parameter to clearly show both covalent bonds and noncovalent interactions.

Most existing analysis tools focus on molecular systems and provide limited support for periodic solid materials. Even programs that support noncovalent interaction analysis for solid materials provide only a limited number of analysis methods. In the study of noncovalent interactions in solid materials, there is currently a

lack of an analysis tool that integrates multiple analysis methods and supports various charge density formats. Based on this background, we developed WIANA for comprehensive analysis of noncovalent interactions. The program supports charge density data calculated by mainstream first-principles software, such as VASP and ABINIT. It only requires charge density data as input and is designed primarily for solid materials. The tool currently integrates three charge-density-based analysis methods: RDG, IRI, and DORI. These methods can be used to observe the distribution of noncovalent interactions, distinguish their types and strengths, and precisely locate interaction regions, helping users quickly identify and further analyze noncovalent interactions in a system from different perspectives.

2. Installation

The program was developed in MATLAB 2022a and is distributed as compiled executable files. Users do not need to install the complete MATLAB environment; the required free MATLAB Runtime toolkit will be installed automatically by the installer. The source code will be released once the program is further refined.

Currently, two versions of the executable are provided:

(1) WIANA_b1_Installer_win_web.exe

Description: MATLAB Compiler Standalone Application Installer for Windows

Tested On: Windows 10 (64-bit)

Executable Path: <Installation_Path>\application\WIANA_b1.exe

(2) WIANA_b1_linux_web.install

Description: MATLAB Compiler Standalone Application Installer for Linux

Tested On: Ubuntu 20.04 LTS (64-bit)

Executable Path: <Installation_Path>/application/WIANA_b1

3. Running the Program

The files required to run the program are the WIANA executable, the input file wiana.ini, and charge density data in CHGCAR, cube, or xsf format. Place these three files in the same directory. No additional configuration is required. Double-click the WIANA executable to start the program and perform the calculation.

3.1 Input Parameters

Calculation parameters are stored in the input file named wiana.ini, which is

automatically read when the program starts. Each line follows the format "Parameter name = Parameter value". Lines starting with # or blank lines are ignored. Parameter values can be numbers, strings, or multiple values separated by commas or spaces.

3.1.1 Sample input file

```
This is an example for the WIANA control file wiana.ini.

FilePath = PATH\CHGCAR                    # Charge density path

CalcType = RDG, IRI, DORI                 # Calculation methods

AlgoType = 1                              # Algorithm: direct

Results_Output = PATH\CHGCAR\result       # Output path

SignRhoMin = -inf                         # No range filtering

SignRhoMax = inf

SaveScatter = 1                           # Save scatter plot
```

3.1.2 Required Parameters

The following parameters must be specified in the configuration file; otherwise, the program will report an error.

(1) FilePath
Type: String
Default: None
Description: Specifies the path to the charge density file. Both absolute and relative paths are supported. The program automatically determines the data format from the filename (e.g., CHGCAR, XSF, CUBE). There is no default value for FilePath.
Example: FilePath = E:\charge_density\CHGCAR.

3.1.3 Optional Parameters

The program provides reasonable default values. Users can adjust them according to calculation needs.

(1) CalcType
Type: String
Default: RDG
Description: Selects the calculation type. Multiple methods can be calculated simultaneously, separated by commas or spaces. Supported functions include: RDG

(Reduced Density Gradient), IRI (Interaction Region Indicator), DORI (Density Overlap Regions Indicator).
Example: CalcType = RDG, IRI, DORI

(2) AlgoType
Type: Integer
Default: 1
Description: Specifies the algorithm used to calculate the charge density gradient. Algorithm 1 calculates the gradient directly on the original fractional-coordinate grid. Algorithm 2 first interpolates the charge density onto a uniform grid and then calculates the gradient. The direct method is faster and preserves the original grid, whereas the interpolation method provides higher accuracy but requires more computation time.
Example: AlgoType = 1

(3) Results_Output
Type: String
Default: Results
Description: Specifies the location where the results are saved. If a relative path is given, the program creates the output folder under the directory containing the configuration file. If an absolute path is given (e.g., E:\data\output), the program creates the output subfolder under the specified path. The naming rule is {CalcType}_{AlgoStr}_{FileType}_{MatName} (e.g., RDG_Direct_VASP_KBBF). If the subfolder already exists, suffixes such as _1 and _2 are automatically added to avoid overwriting existing results.
Example: Results_Output = Results
(4) Pad
Type: Integer
Default: 2
Description: When calculating the charge density gradient, the charge density is periodically extended to reduce errors caused by boundary effects. The default value of 2 is suitable for most systems. Users can adjust it according to actual needs.
Example: Pad = 2

(5) Res_x, Res_y, Res_z
Type: Integer
Default: Res_x = 150, Res_y = 150, Res_z = 150
Description: These parameters are used only for the interpolation method and define the number of uniform grid points in the three directions. The total number of grid

points is Res_x * Res_y * Res_z. Calculation time and memory usage are proportional to the number of grid points. Users can adjust these parameters according to the system size and available computational resources.
Example: Res_x = 150, Res_y = 150, Res_z = 150

(6) ClampMax
Type: Float
Default: 100
Description: This parameter specifies the constant value used to replace unstable or invalid RDG/IRI function values. In RDG and IRI calculations, the charge density near atomic nuclei is relatively high, and the computed function values in this region may become numerically unstable, interfering with the identification and visualization of weak interaction regions. To avoid this problem, for grid points where the charge density exceeds the cutoff threshold, the program does not use their computed RDG/IRI values; instead, it sets the function values directly to ClampMax. In addition, during numerical calculations, some grid points may produce invalid values (NaN) due to near-zero gradients or low density, and the program also replaces these invalid values with ClampMax.
Example: ClampMax = 100

(7) ColorMin / ColorMax
Type: Float
Default: -0.05 / 0.05
Description: These two parameters control the color change range in isosurfaces and scatter plots. The color map typically uses the B-G-R mode (blue-green-red gradient). When the value of $\mathrm{sign}(\lambda_2)\rho$ is equal to or lower than ColorMin, it is shown as blue. When equal to or higher than ColorMax, it is shown as red. These parameters only affect the color display and do not change the x-axis range of the scatter plot. To adjust the x-axis display range, use parameters SignRhoMin and SignRhoMax.
Example: ColorMin = -0.05, ColorMax = 0.05

(8) SignRhoMin / SignRhoMax
Type: Float
Default: -inf / inf (meaning no filtering is applied)
Description: These two parameters specify the range of $\mathrm{sign}(\lambda_2)\rho$. When both parameters are set to finite values, the program keeps only data points within this range for analysis and output. This allows users to focus on specific interaction regions. The filtering affects the scatter plot x-axis range, the output function value files, and the

scatter plot data. When both are set to finite values, the program enables filtering mode, which specifically affects the following:

1. Scatter plot x-axis range: The program automatically adjusts the x-axis display range to [SignRhoMin, SignRhoMax] for easier observation of distribution features within that interval. When filtering is not enabled, the x-axis range uses the recommended ranges from the literature (e.g., [-0.05, 0.05] for RDG).

2. Output grid file value filtering: When exporting visualization data files, the program retains only function values for grid points whose $\mathrm{sign}(\lambda_2)\rho$ values fall within the [SignRhoMin, SignRhoMax] interval. For points outside the interval, the real-space function values (RDG, IRI) are uniformly replaced with ClampMax, and DORI is set to 0. Thus, when drawing isosurfaces in software like VESTA, only the target interaction regions will display isosurfaces, while other regions are effectively masked, avoiding interference from irrelevant isosurfaces.

3. Scatter plot data file filtering output: If parameter SaveData is also enabled, the output text file contains only data points whose $\mathrm{sign}(\lambda_2)\rho$ values are within the specified interval. The file name automatically includes the filtering range, e.g., {MaterialName}_{CalcType}_(SignRhoMin, SignRhoMax).dat.

Example: SignRhoMin = -0.03; SignRhoMax = 0.01

(9) SaveData

Type: Integer (0 or 1)

Default: 1

Description: This parameter controls whether to export data for drawing scatter plots. Set to 1 to export, or 0 to not export. The exported scatter plot raw data file contains three columns:

1. $\mathrm{sign}(\lambda_2)\rho$ value (original, unclipped)

2. Real-space function value (RDG/IRI/DORI)

3. Clipped $\mathrm{sign}(\lambda_2)\rho$ (controlled by ColorMin/ColorMax)

File naming rules:

- Default mode:{MaterialName}_{CalcType}_Full_SignRho.dat
- Filtering mode:{MaterialName}_{CalcType}_(SignRhoMin_to_SignRhoMax).dat

Example: SaveData = 1

(10) SaveScatter

Type: Integer (0 or 1)

Default: 1

Description: This parameter controls whether scatter plot images are exported. Set to

1 to export, or 0 to not export. The exported images are PNG files. When set to 0, the program skips scatter plot drawing and PNG saving. This setting is suitable when available memory is limited, plotting takes too long, or too many points are drawn in the scatter plot, which may cause the program to be terminated by the system.
Example: SaveScatter = 1

(11) RhoMin
Type: Float
Default: 1e-4
Description: Threshold for filtering very low charge density. When the electron density is below RhoMin, the real-space function value at that point is replaced with a specified constant: RDG/IRI are replaced with ClampMax, DORI is replaced with 0. This avoids numerical noise in vacuum or very low-density regions.
Example: RhoMin = 1e-4

(12) IRIexp
Type: Float
Default: 1.1
Description: The IRI method is an extension of the RDG method. It introduces an adjustable empirical parameter a (default 1.1) to enhance discrimination between covalent and noncovalent interaction regions, allowing simultaneous display of various types of interactions.
Example: IRIexp = 1.1

3.2 Input Files

Running the program requires two types of input files:

1. **Input file** (default name wiana.ini): Contains the parameters for the calculation.
2. **Charge density file**: Supports three formats. The program automatically recognizes the format based on the file name or extension:

- **CHGCAR**: Charge density file output by VASP software, usually named CHGCAR or with a material name prefix (e.g., CHGCAR_material).
- **XSF**: Charge density format output by ABINIT, QE, etc., with file extension .xsf.
- **CUBE**: Charge density data format output by Gaussian and other software, with file extension .cube or .cub.

3.3 Output Files

All results are saved in an automatically generated folder. The output path can be changed by modifying the Results_Output parameter. The folder naming rule is:

{CalcType}_{AlgoStr}_{FileType}_{MatName}, CalcType denotes the calculation type(s), and multiple types are connected using plus signs (e.g., RDG + IRI). AlgoStr is set to Direct for the direct-difference algorithm (AlgoType = 1) and to Interp for the interpolation algorithm (AlgoType = 2). FileType is the input file type, and MatName is the material name extracted from the input file name. (e.g., RDG_Direct_VASP_KBBF) If the folder already exists, suffixes like _1, _2 are automatically added.

3.3.1 Output Folder Structure

Taking a KBBF material with charge density output in CHGCAR format (named CHGCAR_KBBF) as an example, analyzing the material with the RDG method results in the following output structure:

```
Results_Output
└── RDG_Direct_VASP_KBBF
        ├──KBBF_RDG.vasp                  # RDG data
        ├──KBBF_RDG_Full_SignRho.dat      # Scatter plot data
        ├──KBBF _sign_rho.vasp            # sign(λ₂)ρ function data
        ├──RDG_Scatter.png                # RDG scatter plot
        └──wiana.log                      # Run log
```

3.3.2 Description of Generated Files

After the program finishes running, all output results are saved in the automatically generated folder. Users can find five types of files in this folder: real-space function values, $\mathrm{sign}(\lambda_2)\rho$ data, scatter plot data, scatter plots, and the run log.

1. **The Real-space function value file (format: .vasp / .cube / .xsf)**. Store the calculated (RDG, IRI, or DORI) data. The file can be directly opened with VESTA, XCrySDen, etc., to draw isosurfaces.
   Naming: MatName_CalcType.extension (e.g., KBBF_RDG.vasp).
2. **$\mathrm{sign}(\lambda_2)\rho$ file (format: .vasp / .cube / .xsf).** Stores the $\mathrm{sign}(\lambda_2)\rho$ value at each grid point. Used to color isosurfaces to distinguish attractive, van der Waals, and repulsive interactions.
   Naming: MatName_sign_rho.extension (e.g., KBBF_sign_rho.vasp).
3. **Scatter plot data file (format: .dat).** Contains three columns: $\mathrm{sign}(\lambda_2)\rho$ data, function value, and clipped $\mathrm{sign}(\lambda_2)\rho$ data. Can be used to recreate scatter plots with software like Origin.
   Naming: MatName_CalcType_Full_SignRho.dat.

If filtering mode is enabled, the name becomes MatName_CalcType_(SignRhoMin_to_SignRhoMax).dat (e.g., KBBF_RDG_(-0.03_to_0.01).dat).

4. **Scatter plot (format: .png).** A 2D scatter plot of $\mathrm{sign}(\lambda_2)\rho$ versus function value automatically generated by the program, using a B-G-R color mapping (blue-green-red gradient).
   Naming: CalcType_Scatter.png.
5. **Run log file (.log)**. Records the parameter settings, calculation progress, time taken, and possible error messages during program execution. This file can be used to check the run status and debug issues.
   Naming: wiana.log.

4. Program Implementation

The implementation workflow of WIANA is shown in Figure 1. The program is mainly divided into three parts: data reading and processing, numerical calculation, and result output and visualization. This chapter describes how each module works and how the data is processed.

4.1 Charge Density Reading Module

The program identifies the file format from the file extension or keywords in the file. It then extracts the lattice vectors, atomic coordinates, and 3D charge density data. After reading the file, the program converts the relevant geometric quantities and charge density to atomic units. This ensures consistent units in the following calculations.

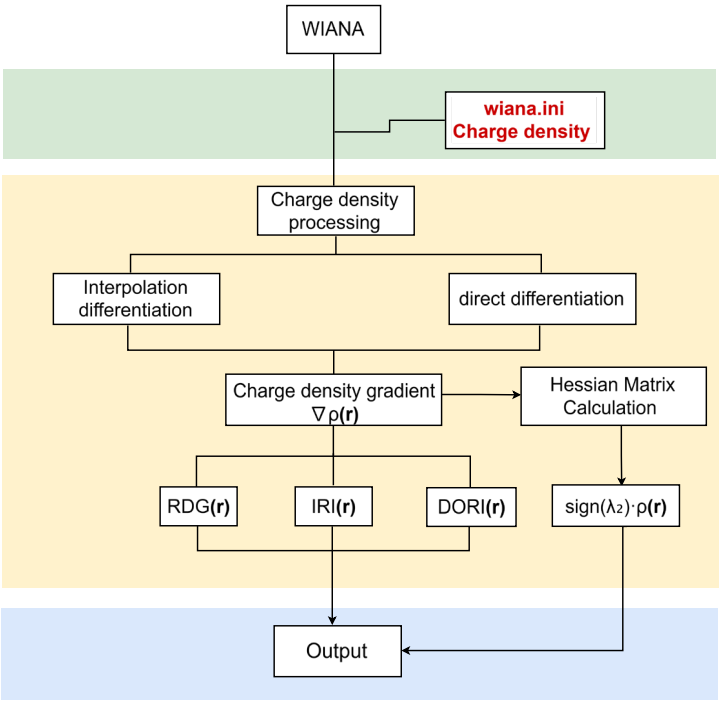


Figure 1 Program Flowchart

1. CHGCAR format. The CHGCAR format stores the charge density generated by VASP after a self-consistent calculation. The program identifies CHGCAR files by the keyword CHGCAR in the file name or by the .vasp extension. When reading the file, the program extracts the lattice vectors, atomic species, atomic coordinates, 3D grid dimensions (NGX, NGY, NGZ), and charge density at each grid point. For subsequent calculations, the program converts the lattice vectors to Bohr and the charge density to atomic units ($1/Bohr^3$).

2. XSF format. The XSF format is a charge density data format output by software such as Quantum ESPRESSO (QE) and ABINIT. The program identifies it by the .xsf extension. The program reads lattice vectors from the PRIMVEC block and converts them to Bohr, reads atomic coordinates from the PRIMCOORD block, and reads grid dimensions and charge density values from the BEGIN_BLOCK_DATAGRID_3D block. The charge density in XSF files is already in atomic units, so no unit conversion is needed after reading.

3. CUBE format. The CUBE format is a general 3D grid data format used by quantum chemistry software such as Gaussian. The program identifies it by the extension .cube or .cub. The header contains the number of atoms, grid origin, number of grid points in each direction, and step vectors between adjacent grid points. All distances are given in Bohr. The density data is stored in Z→Y→X order, layer by layer. After reading the file, the program changes the order to X→Y→Z for subsequent calculations.

4.2 Numerical Calculation Module

This module calculates the charge density gradient ($\nabla\rho$), its magnitude $|\nabla\rho|$, and the eigenvalues of the Hessian matrix. It then calculates $\mathrm{sign}(\lambda_2)\rho$, which is used to distinguish different interaction types. The program provides two numerical algorithms, which users can select via the parameter AlgoType.

4.2.1 Direct Method (AlgoType = 1)

The direct method calculates the charge density gradient in fractional coordinates. It is suitable for periodic crystals and molecular systems and runs relatively fast.

1. **Periodic extension**. At boundary grid points, the gradient function uses a less accurate one-sided difference because there are no neighboring points outside the boundary. To avoid this, the program applies periodic padding to the charge density. This allows boundary grid points to use the more accurate central difference. The number of extended boundary layers is specified by the parameter Pad. The program uses periodic replication:

   `rho_pad = padarray(rho, [Pad, Pad, Pad], 'circular').`

2. **Calculate charge density gradient**. On the padded grid, the program uses the gradient function to calculate the first derivative of the charge density. In fractional coordinates, the step sizes are set to 1/Nx, 1/Ny, and 1/Nz, respectively. This gives the fractional gradient components gx_c, gy_c, and gz_c:

   `[gx_c, gy_c, gz_c] = gradient(rho_pad, 1/Nx, 1/Ny, 1/Nz)`

3. **Coordinate transformation**. The gradient components in fractional coordinates are then transformed to Cartesian coordinates. The transformation formula is:

$$\mathrm{g_cart} = (\mathrm{M}^{-1})^{\mathrm{T}} \cdot \mathrm{g_frac}$$

   This gives the Cartesian gradient components gx, gy, gz and their magnitude.

4. **Construct Hessian matrix and extract λ_2**. To obtain $\mathrm{sign}(\lambda_2)\rho$, the second derivative of the density is needed. The program extends the fractional-coordinate gradient again. It then calculates the second derivatives in each direction and constructs a 3 × 3 Hessian matrix H_cry in fractional coordinates. After symmetrizing this matrix, it transforms the Hessian matrix from fractional coordinates to Cartesian coordinates via similarity transformation:

$$\mathrm{H_cart} = (\mathrm{M}^{-1})^{\mathrm{T}} \cdot \mathrm{H_cry} \cdot \mathrm{M}^{-1}$$

   Finally, the eigenvalues of H_cart are computed and sorted in ascending order, with the second eigenvalue taken as λ_2 The program then multiplies the sign of λ_2 by the charge density to obtain $\mathrm{sign}(\lambda_2)\rho$.

4.2.2 Interpolation Method (AlgoType = 2)

This method first interpolates the charge density onto a user-defined uniform Cartesian grid. The gradient is then calculated on the uniform grid, and the results are interpolated again. It is suitable for periodic crystals and molecular systems. The interpolation method provides higher accuracy but requires more computation time.

1. Periodic boundary extension. The program periodically extends the original charge density grid in the three directions. The number of extended boundary layers is specified by the parameter Pad (default 2).

2. Interpolation to uniform grid. The program generates a uniform grid according to Res_x, Res_y, and Res_z. It uses the griddata function with linear interpolation to map the extended density data onto this grid.

3. Charge Density Gradient and Hessian Calculation. The program directly computes the charge density gradient and Hessian matrix in Cartesian coordinates on the uniform grid. The calculation of $\mathrm{sign}(\lambda_2)\rho$ is the same as in the direct method: the second eigenvalue of the Hessian matrix is taken as λ_2, and its sign is multiplied by the charge density to obtain the $\mathrm{sign}(\lambda_2)\rho$ function.

4.3 Function Calculation Module

Based on the charge density gradient magnitude $|\nabla\rho|$ and charge density ρ, the program calculates the selected function: reduced density gradient (RDG), interaction region indicator (IRI), or density overlap regions indicator (DORI).

4.3.1 Reduced Density Gradient (RDG)

RDG describes how much the charge density distribution deviates from a uniform state. Its definition is:

$$\mathrm{RDG}(\mathrm{r}) = \frac{1}{2(3\pi^2)^{1/3}} \frac{|\nabla\rho(\mathrm{r})|}{\rho(\mathrm{r})^{4/3}}$$

Implementation details:

1. Density cutoff: When the charge density ρ is high, the gradient magnitude $|\nabla\rho|$ may be small or even approach zero, and dividing one by the other can produce unphysical extreme values, thereby interfering with the identification of weak interaction regions. Therefore, for grid points with charge density greater than 0.05 a.u., the program does not calculate the RDG value; it sets the RDG value at such grid points directly to a large constant, i.e., ClampMax (default: 100). This constant can be adjusted through the ClampMax parameter. When filtering mode is enabled, the actual cutoff threshold is set to the larger of |SignRhoMin| and |SignRhoMax|.

2. Invalid and low-density handling: Invalid values (NaN) generated

during calculation are also replaced with ClampMax. In addition, when the charge density is below RhoMin (default: 1e-4), the RDG value is also set to ClampMax to avoid numerical noise in vacuum regions.

4.3.2 Interaction Region Indicator (IRI)

IRI is an improved version of the RDG method. It enhances the ability to distinguish covalent bond regions from weak interaction regions by introducing an adjustable empirical parameter a. Its definition is:

$$\mathrm{IRI(r)} = \frac{|\nabla \rho(\mathrm{r})|}{|\rho(\mathrm{r})|^{\mathrm{a}}}$$

Implementation details:

1. **Adjustable empirical parameter**: The parameter a is controlled by the parameter IRIexp (default 1.1).
2. **Density cutoff**: Similar to RDG, the program applies a cutoff to high-density regions. When the charge density is greater than 0.5 a.u., the IRI value is set to ClampMax. When filtering mode is enabled, the actual cutoff threshold is set to the larger of |SignRhoMin| and |SignRhoMax|.
3. **Invalid and low-density handling**: Invalid values (NaN) generated during calculation are replaced with ClampMax. For low-density filtering (ρ < RhoMin), the value is also set to ClampMax.

4.3.3 Density Overlap Regions Indicator (DORI)

DORI can show both covalent bonds and weak interaction regions by detecting deviations of the charge density from a single exponential decay. Its definition is:

$$\mathrm{DORI}(r) = \frac{\theta(r)}{1+\theta(r)} \qquad \theta(r) = \frac{\left\{\nabla\left[\frac{\nabla\rho(r)}{\rho(r)}\right]^2\right\}^2}{\left[\frac{\nabla\rho(r)}{\rho(r)}\right]^6}$$

Implementation details:

1. **Numerical stability handling**: To avoid numerical issues caused by a zero or very small denominator, a small constant EPS_DORI (default: 1e-12) is introduced in the denominator.
2. **Invalid and low-density handling**: DORI ranges from 0 to 1 and does not require a density cutoff. Invalid values (NaN) are replaced with 0. For low-density filtering (ρ < RhoMin), the value is also set to 0.

4.4 Data Output Module

After the calculation is completed, the program automatically saves the results in the output folder. These results include function value data, scatter plot data, scatter plot images, and a run log. This section describes how these files are generated and processed.

4.4.1 Function Value Data Output

The program generates data files in the same format as the input file (.vasp, .cube, or .xsf). These files can be opened directly with software such as VESTA or XCrySDen to draw isosurfaces. Each calculation type generates two files. The first file stores the real-space function value and is named {MaterialName}_{CalcType}.extension. (e.g., KBBF_RDG.vasp) It is used to determine the spatial location of the isosurface. The second file stores the clipped $\mathrm{sign}(\lambda_2)\rho$ value and is named MaterialName_sign_rho.extension. (e.g., KBBF_sign_rho.vasp) It is used to color the isosurface and distinguish different interaction types. The clipping of $\mathrm{sign}(\lambda_2)\rho$ is controlled by ColorMin and ColorMax. Values outside this range are set to the corresponding boundary value.

4.4.2 Scatter Plot Data Output

When parameter SaveData is set to 1 (the default), the program outputs a three-column text file. Users can import this file into software such as Origin to create or modify scatter plots. Set parameter SaveData to 0 to disable this output. For each valid grid point, the file records three values in the following order:

1. **Original $\mathrm{sign}(\lambda_2)\rho$ value.**
2. **Function value:** the real-space function value calculated by WIANA (RDG, IRI, or DORI).
3. **Clipped $\mathrm{sign}(\lambda_2)\rho$ value:** the value after applying the ColorMin and ColorMax limits. This value is used for color mapping.

The default file name is {MaterialName}_{CalcType}_Full_SignRho.dat. When filtering mode is enabled, the file name includes the filtering range, for example, {MaterialName}_{CalcType}_(SignRhoMin_to_SignRhoMax).dat. Only data points within the specified range are saved.

4.4.3 Automatically Generated Scatter Plot

The program can automatically generate scatter plots. For each calculation type, the program generates a scatter plot with $\mathrm{sign}(\lambda_2)\rho$ on the horizontal axis and the function value on the vertical axis. The plot is saved as a PNG file named {CalcType}_Scatter.png. The scatter points use a B-G-R color scheme (blue-green-red gradient). The color range is controlled by ColorMin and ColorMax, showing the

distribution of attractive, van der Waals, and repulsive interactions. The axis ranges use the literature-recommended values (RDG: x-axis [-0.05, 0.05], y-axis [0, 2.0]; IRI: x-axis [-0.4, 0.4], y-axis [0, 5.0]; DORI: x-axis [-0.4, 0.4], y-axis [0, 1]). When filtering is enabled, the x-axis automatically adapts to the user-specified interval from SignRhoMin to SignRhoMax.

4.4.4 Run Log File

During program execution, the program writes key information to wiana.log, including input parameters, grid information, calculation progress, the number of valid points, function value ranges, and run time.

5. Calculation Examples

5.1 Example 1: Phenol dimer

To verify the correctness and reliability of this program in handling noncovalent interactions, we selected the phenol dimer as a benchmark test. This system simultaneously contains hydrogen bonds, steric repulsion, and van der Waals interactions in one configuration, making it an ideal test model. The calculation results have been verified by comparison with those from Critic2 and Multiwfn.

The parameters in the input file wiana.ini are set as follows:

```
FilePath = E:\Matlab\CHGCAR_phenol-dimer
CalcType = RDG IRI
AlgoType = 1
Results_Output = Output
#Res_x = 150
#Res_y = 150
#Res_z = 150
IRIexp = 1.1
ColorMin = -0.05
ColorMax = 0.05
SignRhoMin = -inf
SignRhoMax = inf
ClampMax = 100
```

```
RhoMin = 1e-4
SaveScatter = 1
SaveData =1
```

The generated files are stored in the Output folder, as shown in Figure 2.

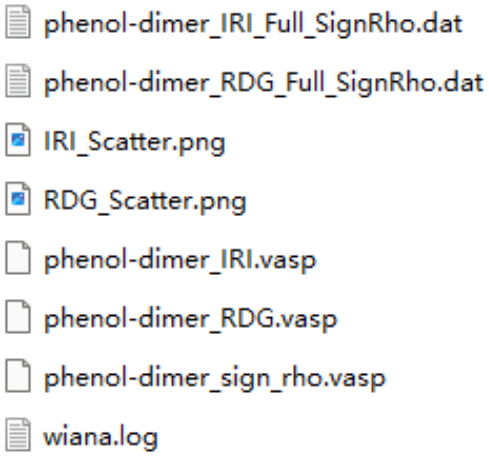


Figure 2 Output files generated by the program

The scatter plot generated by the program is shown in Figure 3.

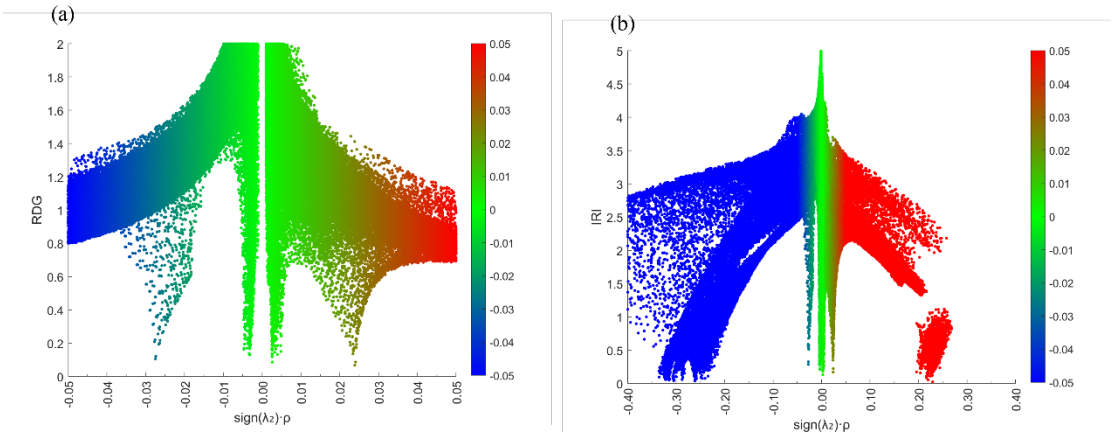


Figure 3 Phenol dimer NCI plots: (a) RDG vs. $\mathrm{sign}(\lambda_2)\rho$ scatter plot; (b) IRI vs. $\mathrm{sign}(\lambda_2)\rho$ scatter plot

The steps for drawing an isosurface using VESTA are as follows (taking the RDG isosurface as an example):

1. Open VESTA and drag the RDG data file generated by the program (phenol-dimer_RDG.vasp) into VESTA.

2. From the menu bar, select Edit > Edit Data > Volumetric Data. Click Import next to Isosurface and Surface coloring, respectively, to import the RDG data and the $\mathrm{sign}(\lambda_2)\rho$ data, as shown in Figure 4.

3. From the menu bar, select Objects > Properties > Isosurfaces. Set the Isosurface level to 0.5. For Surface coloring, the default Max/Min values of Saturation level are [0.05, -0.05]. These values can be adjusted as needed, as

shown in Figure 5.

4. Adjust other display settings: uncheck Objects > Volumetric Data > Show Sections.

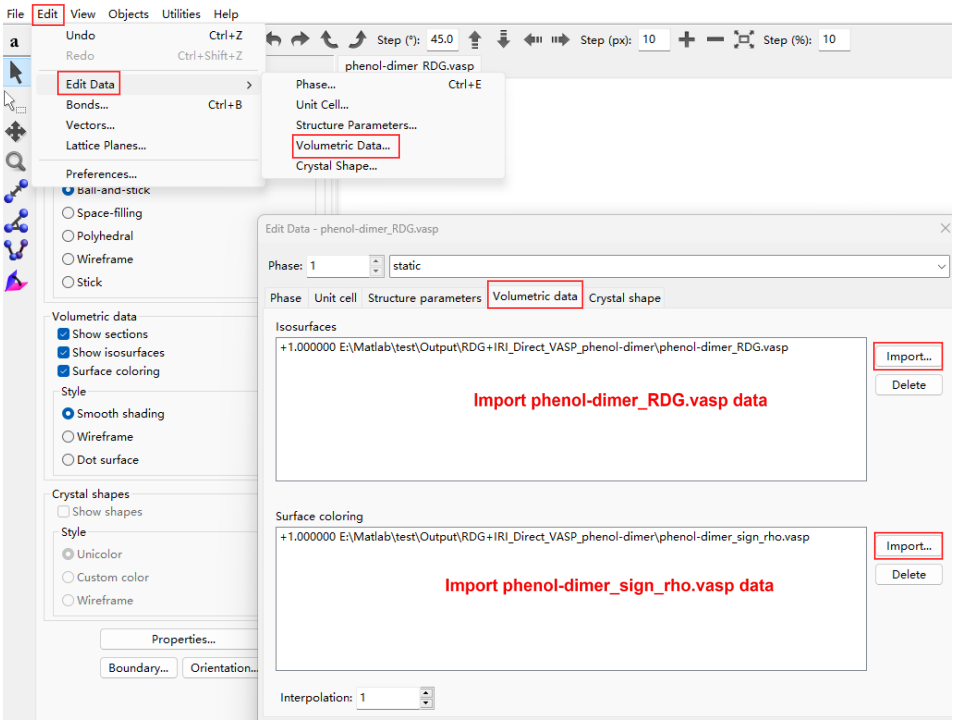


Figure 4 VESTA isosurface drawing step 1

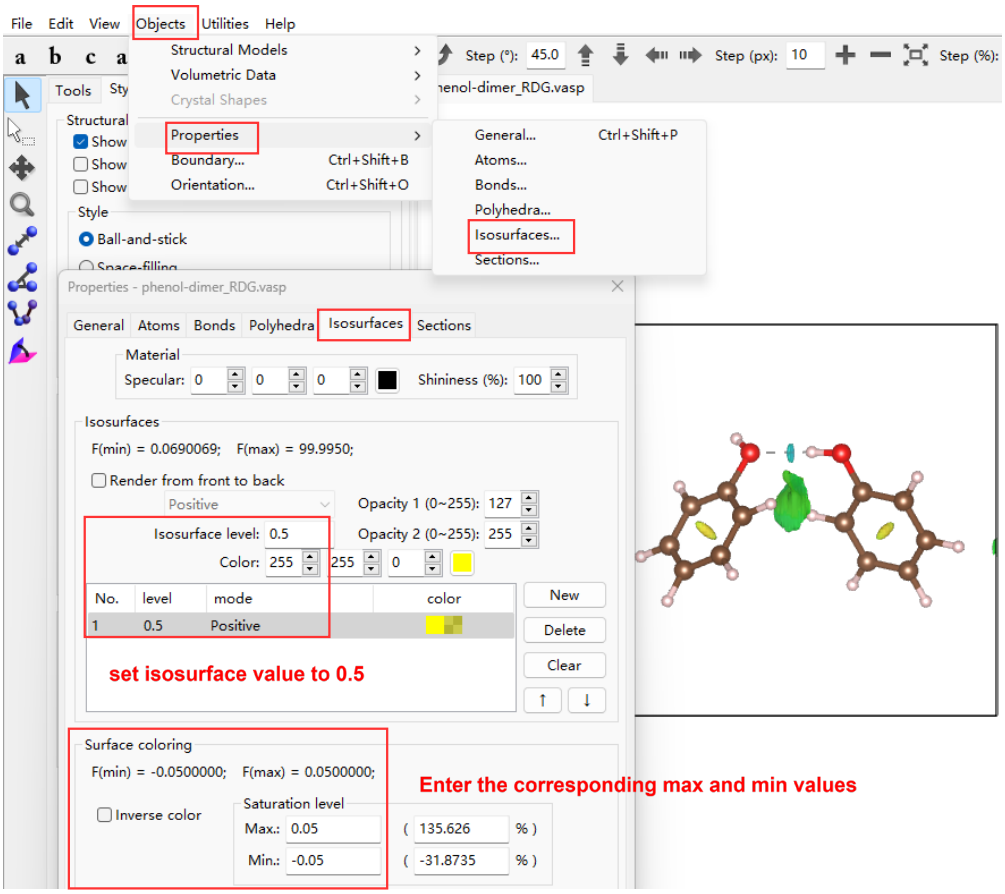


Figure 5 VESTA isosurface drawing step 2

The steps for drawing the IRI and DORI isosurfaces are the same as those described above. The IRI isosurface is usually set to 1.1, and the DORI isosurface is set to 0.99. The RDG and IRI isosurfaces are shown in Figure 6.

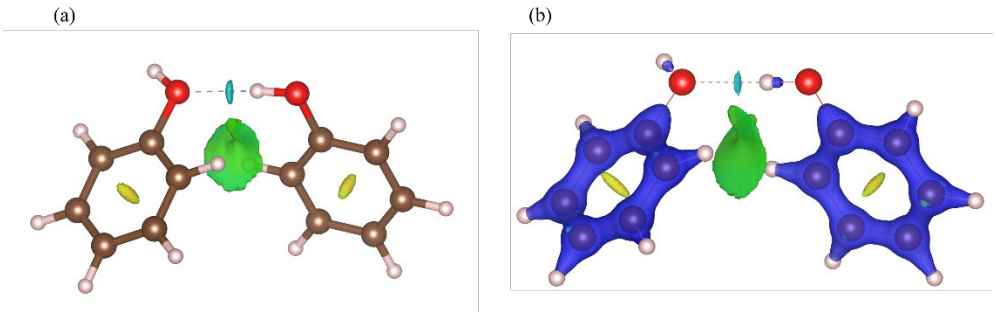


Figure 6 Phenol dimer isosurface plots: (a) RDG = 0.5 isosurface; (b) IRI = 1.1 isosurface.

Figure 7 shows the calculation results of the phenol dimer using the RDG and IRI methods. It can be seen that the calculation results are basically consistent with those reported in the literature. Figure 7(a) is the scatter plot of the phenol dimer obtained with the RDG method, and Figure 7(b) is the scatter plot obtained with the IRI method . In the region below 0.05, the calculation results obtained using the IRI method are similar to those obtained using the RDG method. The peak appearing in the region above 0.05 corresponds to stronger interactions. Figure 7(c) corresponds to the

isosurface plot with RDG = 0.5 and -0.05 < $\text{sign}(\lambda_2)\rho$< 0.05. It can be seen that when RDG = 0.5, there are four spikes. The corresponding hydrogen bond, steric repulsion, and van der Waals interactions can be found in this isosurface plot, shown in blue, red, and green, respectively. Figure 7(d) corresponds to the isosurface plot with IRI = 1.1 and -0.4 < $\text{sign}(\lambda_2)\rho$< 0.4, which also shows covalent interactions between atoms.

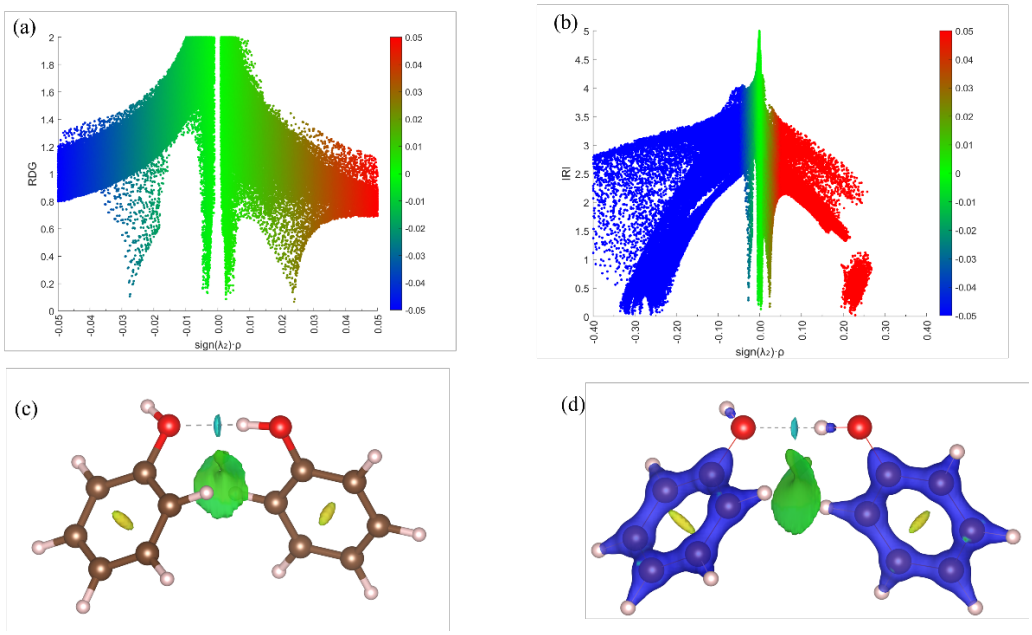


Figure 7 NCI and IRI plots for phenol dimer: (a) RDG vs. $\text{sign}(\lambda_2)\rho$ scatter plot; (b) IRI vs. $\text{sign}(\lambda_2)\rho$ scatter plot; (c) RDG = 0.5 isosurface colored by $\text{sign}(\lambda_2)\rho$; (d) IRI = 1.1 isosurface colored by $\text{sign}(\lambda_2)\rho$.

5.2 Example 2: $CH_3NH_3PbI_3$

Organic-inorganic hybrid perovskite materials have great application potential in the optoelectronic field due to their excellent properties. The noncovalent interactions between organic cations and inorganic frameworks, such as hydrogen bonds and halogen bonds, play an important role in regulating material stability and charge transport behavior. In this section, the classic perovskite material $CH_3NH_3PbI_3$ is used as an example to analyze its noncovalent interactions.

The parameters in the input file wiana.ini are set as follows:

```
FilePath = E:\Matlab\CHGCAR_CH3NH3PbI3
CalcType = RDG DORI
AlgoType = 1
Results_Output = Output
#Res_x = 150
#Res_y = 150
```

```
#Res_z = 150
IRIexp = 1.1
ColorMin = -0.05
ColorMax = 0.05
SignRhoMin = -inf
SignRhoMax = inf
ClampMax = 100
RhoMin = 1e-4
SaveScatter = 1
SaveData =1
```

The generated files are stored in the Output folder, as shown in Figure 8.

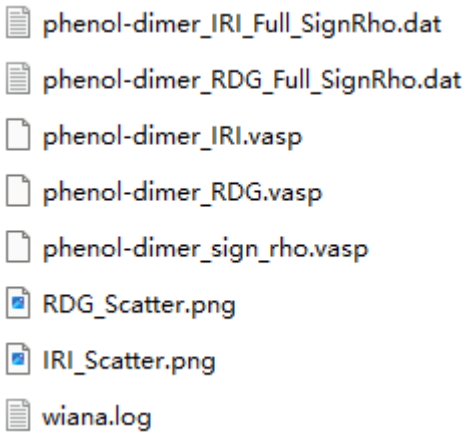


Figure 8 Output files generated by the program

The scatter plots automatically generated by the program for the two analysis methods are shown in Figure 9.

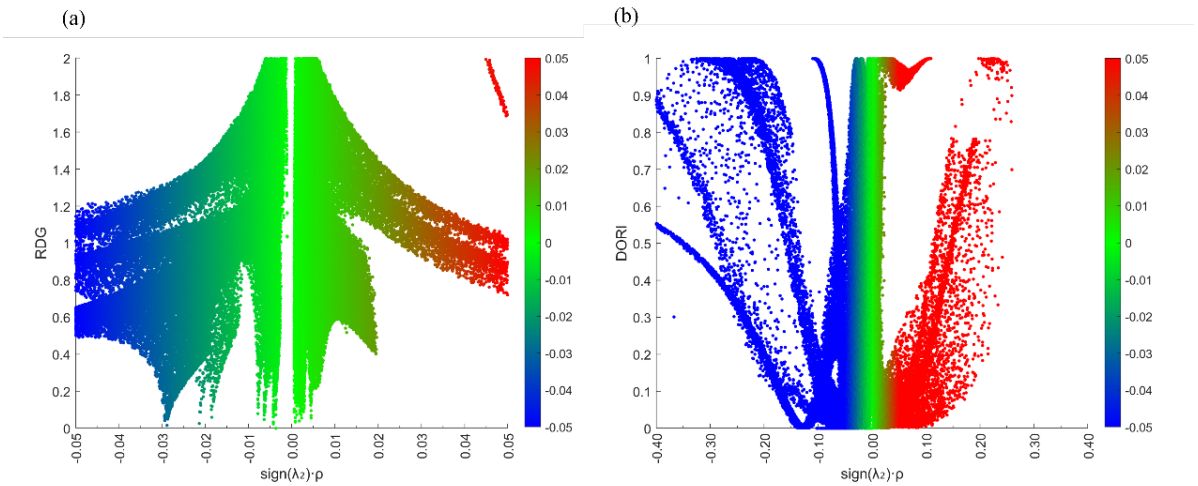


Figure 9 NCI and DORI scatter plots for $CH_3NH_3PbI_3$: (a) RDG vs. $\mathrm{sign}(\lambda_2)\rho$ scatter plot;

(b) DORI vs. $\text{sign}(\lambda_2)\rho$ scatter plot

The isosurface plots are as follows in Figure 10:

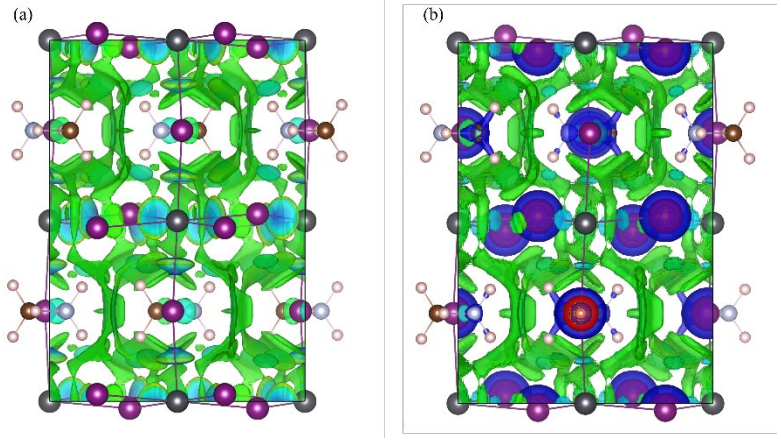


Figure 10 Isosurface plots for $CH_3NH_3PbI_3$: (a) RDG = 0.5 isosurface; (b) DORI = 0.99 isosurface

The structural feature of $CH_3NH_3PbI_3$ is a network of corner-sharing PbI_6 octahedra, with $[CH_3NH_3]^+$ located in the voids surrounded by the octahedra Figure 9(a) presents the RDG scatter plot, which exhibits several characteristic peaks located in the $\text{sign}(\lambda_2)\rho \approx -0.03$, $\text{sign}(\lambda_2)\rho \approx -0.02$, and $-0.01 < \text{sign}(\lambda_2)\rho < 0.01$regions. Compared with the RDG analysis, the DORI scatter plot in Figure 9(b) reveals additional peaks in the $\text{sign}(\lambda_2)\rho \approx -0.1$ and $-0.3 < \text{sign}(\lambda_2)\rho < -0.2$.

To visualize the spatial locations of the interactions more intuitively, isosurface plots are needed. The SignRhoMin and SignRhoMax parameters in the wiana.ini file can be adjusted to restrict the output data to a specified range for drawing isosurfaces.

Set the parameters as follows:

```
SignRhoMin = -0.035
SignRhoMax = -0.025
```

The resulting isosurface is presented in Figure 11(a). The $\text{sign}(\lambda_2)\rho \approx -0.03$ corresponds to the electrostatic interaction between Pb and I atoms in the PbI_6 octahedra.

Modify the SignRhoMin and SignRhoMax parameters as follows:

```
SignRhoMin = -0.025
SignRhoMax = -0.015
```

The resulting isosurface is presented in Figure 11(b). Two sharp peaks appear in the $\text{sign}(\lambda_2)\rho \approx -0.02$, corresponding to the hydrogen bonding between the H atoms bonded to N and the adjacent I atoms, together with the electrostatic interactions between Pb and I atoms in the PbI_6 octahedra.

Set the parameters as follows:

```
SignRhoMin = -0.01
SignRhoMax = 0.00
```

The corresponding isosurface is presented in Figure 11(c). The sharp peak in the $-0.01 < \mathrm{sign}(\lambda_2)\rho < 0.00$ can be attributed to van der Waals interactions, including those between the H atoms bonded to C and the adjacent I atoms, as well as those between I atoms.

Set the parameters as follows:

```
SignRhoMin = 0.00
SignRhoMax = 0.01
```

The resulting isosurface is shown in Figure 11(d). The sharp peak in the $0.00 < \mathrm{sign}(\lambda_2)\rho < 0.01$ originates predominantly from van der Waals interactions between I atoms belonging to different PbI_6 octahedra.

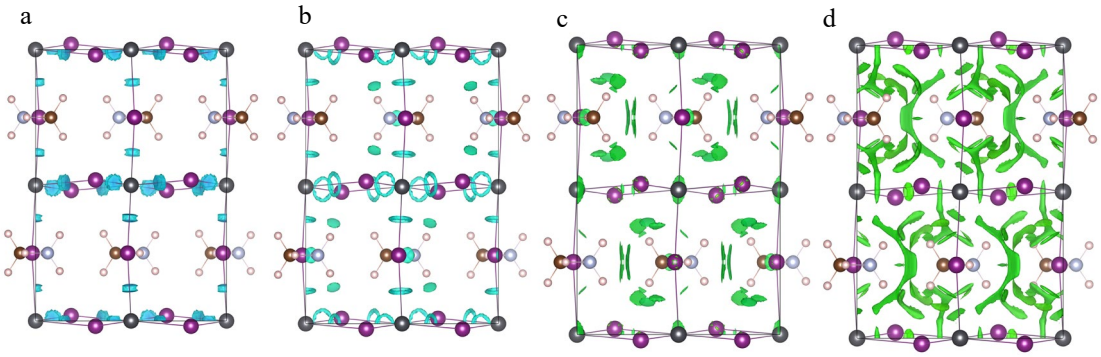


Figure 11 Isosurface maps (RDG = 0.5) of $CH_3NH_3PbI_3$ for different $\mathrm{sign}(\lambda_2)\rho$ ranges: (a) $-0.035 < \mathrm{sign}(\lambda_2)\rho < -0.025$, (b) $-0.025 < \mathrm{sign}(\lambda_2)\rho < -0.015$, (c) $-0.01 < \mathrm{sign}(\lambda_2)\rho < 0.00$, and (d) $0.00 < \mathrm{sign}(\lambda_2)\rho < 0.01$.

5.3 Example 3: $NH_4B_4O_6F$

$NH_4B_4O_6F$ (ABF) is an excellent deep-ultraviolet SHG optical crystal and plays a very important role in all-solid-state lasers. Its structural feature is $[B_4O_6F]_\infty^{-1}$layer, which connects through hydrogen bonds with $[NH_4]^+$. According to the literature, the layer spacing of ABF is 3.81 Å. This key feature gives ABF a relatively high interlayer binding energy, effectively alleviating the significant layered growth characteristics and laying a structural foundation for growing thick ABF crystals. We use the program to calculate and analyze the interlayer interactions.

The parameters in the input file wiana.ini are set as follows:

```
FilePath = E:\Matlab\CHGCAR_NH4B4O6F
CalcType = RDG
AlgoType = 2
Results_Output = Output
```

```
Res_x = 130
Res_y = 160
Res_z = 130
IRIexp = 1.1
ColorMin = -0.05
ColorMax = 0.05
SignRhoMin = -inf
SignRhoMax = inf
ClampMax = 100
RhoMin = 1e-4
SaveScatter = 1
SaveData =1
```

The generated files are stored in the Output folder, as shown in Figure 12.

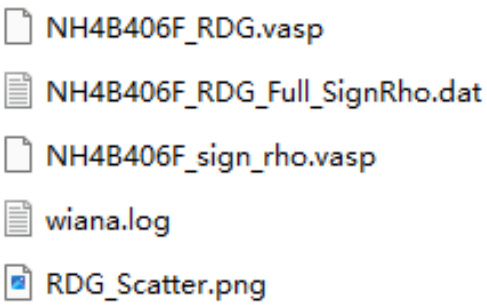


Figure 8 Output files generated by the program

The RDG scatter plot and the RDG = 0.5 isosurface generated by the program are shown in Figure 13.

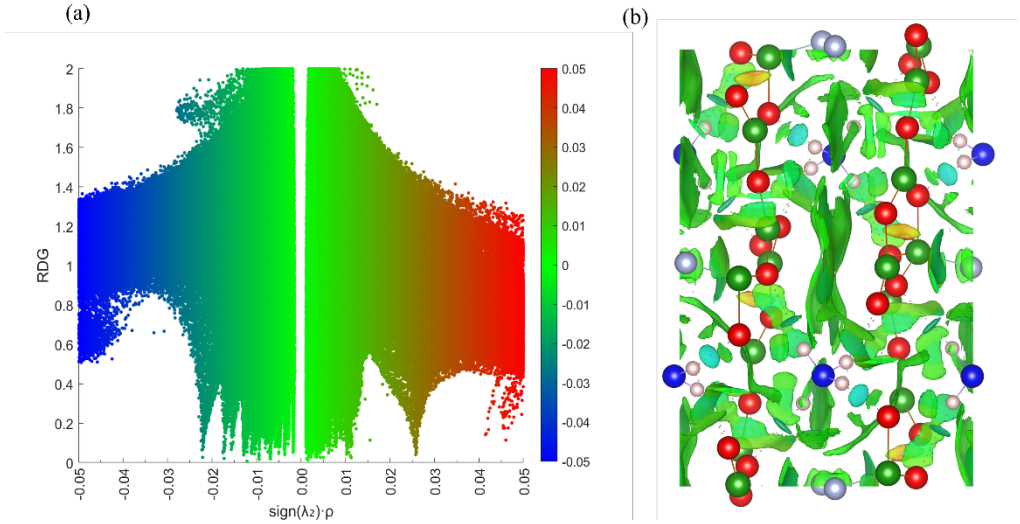


Figure 9 ABF RDG analysis: (a) RDG vs. $\mathrm{sign}(\lambda_2)\rho$ scatter plot; (b) RDG = 0.5 isosurface

As shown in Figure 13(a), a spike appears in the region $\mathrm{sign}(\lambda_2)\rho$ = -0.03, corresponding to hydrogen-bond interactions between H in $[NH_4]^+$ and O in $[B_4O_6F]_\infty^{-1}$. The spike in the region

-0.01 < $\mathrm{sign}(\lambda_2)\rho$< 0.01 corresponds to van der Waals interactions between $[B_4O_6F]_\infty^{-1}$layers and between $[B_4O_6F]_\infty^{-1}$and $[NH_4]^+$. A spike appears in the region $\mathrm{sign}(\lambda_2)\rho \approx 0.023$, corresponding to steric repulsion regions within the B-O rings inside the $[B_4O_6F]_\infty^{-1}$ layer.

To more clearly show the positions of the interactions, modify the two parameters SignRhoMin and SignRhoMax in the wiana.ini file to restrict the output data within this range for isosurface drawing. For example, change these two parameters to:

```
SignRhoMin = -0.03
SignRhoMax = -0.015
```

The resulting isosurface shows only hydrogen bonding interactions, as shown in Figure 14.

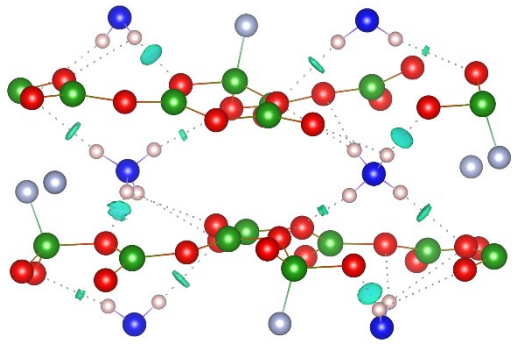


Figure 10 ABF RDG = 0.5 isosurface, $\mathrm{sign}(\lambda_2)\rho$ range: [-0.03, -0.015]

6. Frequently Asked Questions

Q1: The input file cannot be found, or the file name is misspelled.

Hint: The configuration file wiana.ini does not exist or the file name is incorrect. Please confirm that a file named wiana.ini exists in the current working directory.

Q2: The charge density path is incorrect.

Hint: The charge density file does not exist. Please check whether the path specified by the FilePath parameter in the configuration file is correct and whether the corresponding file exists at that path.

Q3: The input file contains unreasonable symbols (such as semicolons, periods, question marks, etc.)

Hint: The configuration file contains extra symbols. Please use an equals sign to

separate parameter names and values, and do not add commas, semicolons, periods, question marks, or other extraneous symbols around parameter values or names.

Q4: Parameter name misspelled.

Hint: The parameter name is misspelled. Please check whether the parameter names appearing in the configuration file are exactly the same as those defined by the program, namely: CalcType, AlgoType, FilePath, Pad, Res_x, Res_y, Res_z, ClampMax, ColorMin, ColorMax, IRIexp, SignRhoMin, SignRhoMax, RhoMin, SaveScatter, SaveData, Results_Output.

Q5: The AlgoType parameter is set to a value other than 1 or 2.

Hint: The AlgoType parameter value is invalid. This parameter can only be 1 (for direct method) or 2 (for interpolation method). Please modify it to the correct value.

Q6: Charge density read error (file format not recognized)

Hint: Unable to automatically recognize the format of the charge density file. Please ensure the file extension is .cube, .cub, or .xsf, or that the file name is exactly CHGCAR (without any extension).

Q7: The program terminates during the scatter plot generation stage.

Hint: When there are too many scatter plot data points, generating the scatter plot consumes a large amount of memory and causes the program to be terminated by the system. To avoid this problem, set the parameter **SaveScatter** to 0 in the **wiana.ini** file to skip image output. It is recommended to set it to 0 when available memory is limited, plotting takes too long, or only data output is required.

7.References

[1] JOHNSON E R, KEINAN S, MORI-SANCHEZ P, et al. Revealing noncovalent interactions [J]. J Am Chem Soc, 2010, 132(18): 6498-6506.

[2] DE SILVA P, CORMINBOEUF C. Simultaneous Visualization of Covalent and Noncovalent Interactions Using Regions of Density Overlap [J]. J Chem Theory Comput, 2014, 10(9): 3745-3756.

[3] LU T, CHEN Q. Interaction Region Indicator: A Simple Real Space Function Clearly Revealing Both Chemical Bonds and Weak Interactions [J]. Chemistry–Methods, 2021, 1(5): 231-239.